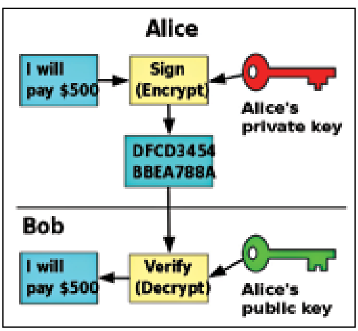
Cryptographic algorithms and security – married by keys

Though we have a chapter dedicated to cryptanalysis, we'll give you quick rundown on what it takes to break a cipher and what factors govern the security of a cryptographic algorithm.

Algorithm strength

A very strong key and a weak algorithm will result in a weak cipher. For convenience sake, let's take the example we showed in the beginning again. We're rotating the alphabet by n number of times. Here n is the key. Now, it won't matter if we put n = 200 or n = 1000, there's a limit to its effectiveness. The effectiveness limit is 25 times. You can't rotate alphabets by more than 25 times. The 26th time it would always come back to the same position as if the rotation was never done. In simple terms, the algorithm itself is limiting the difficulty it can pose to an unintended recipient against cracking the cipher (converting ciphertext back to plaintext).

Not only this, the algorithm is very simple too. Rotating alphabets is something a child can do easily to pass messages to his friends in a classroom. It's no big deal. Consider a scenario where the teacher catches the paper on which he was passing the message. The message is encrypted but how difficult would it be for the teacher to interpret? All she would have to do is put herself in the child's place thinking "what can a child do to encrypt a message?" Add some trial and error into the mix and the message being passed would be easily interpreted by the unintended recipient (the teacher in this case). Hence, for maintaining security, using a good algorithm is important.



Asymmetric algorithms are also used for digital signatures

Computational power required

Okay, so you own a mobile phone that's more powerful than a computer NASA sent to the moon years ago. But is that power enough to break a DES cipher? What about a Triple DES? Or RSA perhaps? Nope, right? Your mobile phone wouldn't be able to do it. If you have the ciphertext and the algorithm that was used, then you need to know